# BAMAT101: Multivariable Calculus & Differential Equations

**Course Module Overview & Structure** | Vellore Institute of Technology

## Module 1: Partial Differentiation

Functions of two or more variables, Limits, Continuity, Partial derivatives, Total differential, Jacobians, Taylor's series for functions of two variables, Maxima and minima, Lagrange's multiplier method.

## Module 2: Multiple Integrals

Double integrals, Change of order of integration, Double integrals in polar coordinates, Area enclosed by plane curves, Triple integrals, Volume of solids, Change of variables (Spheroidal and Cylindrical coordinates).

## Module 3: Vector Differentiation

Scalar and vector point functions, Gradient, Directional derivative, Divergence, Curl, Vector identities, Solenoidal and Irrotational vector fields.

## Module 4: Vector Integration

Line, Surface, and Volume integrals, Green's theorem in a plane, Gauss Divergence theorem, Stokes' theorem (without proofs), Verification and evaluation of vector integrals using these theorems.

## Module 5: First Order Ordinary Differential Equations

Linear and Bernoulli's equations, Exact differential equations, Equations reducible to exact equations, Integrating factors, Orthogonal trajectories, Newton's law of cooling.

## Module 6: Higher Order Linear Ordinary Differential Equations

Linear differential equations of higher order with constant coefficients, Method of variation of parameters, Cauchy-Euler homogeneous equations, Legendre's linear equations, Simultaneous linear equations.

## Module 7: Partial Differential Equations

Formation of partial differential equations, Solutions of standard types of first-order partial differential equations, Lagrange's linear equation, Linear partial differential equations of second and higher order with constant coefficients.